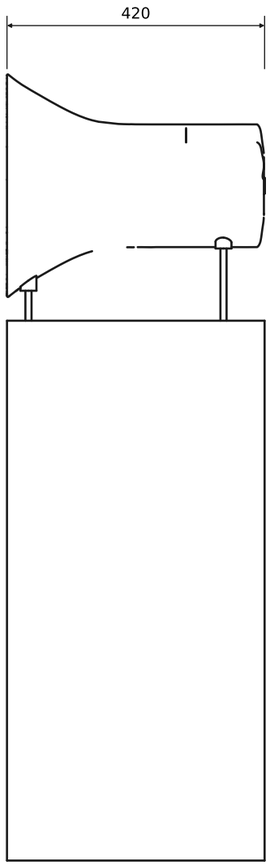
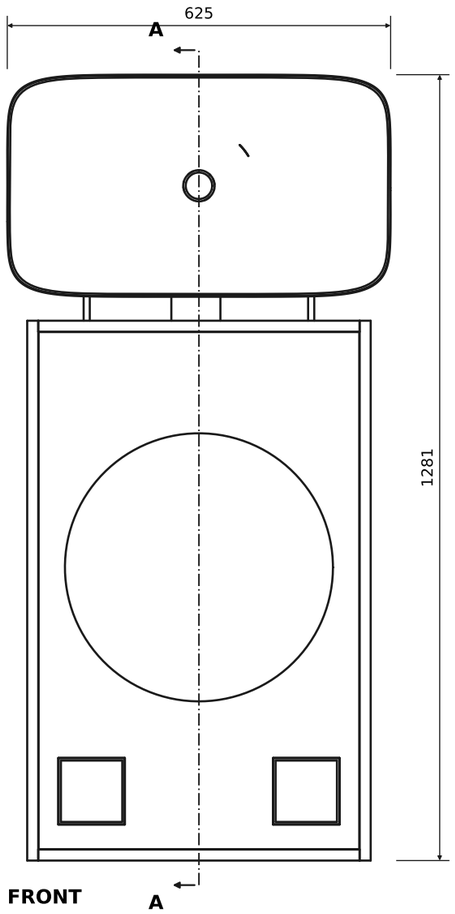
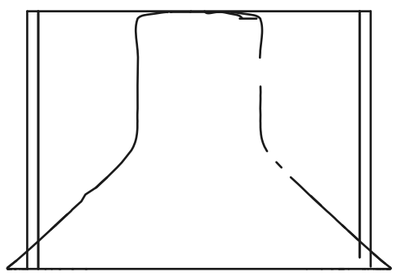
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.




FROM THE RIGHT




FRONT



FROM ABOVE

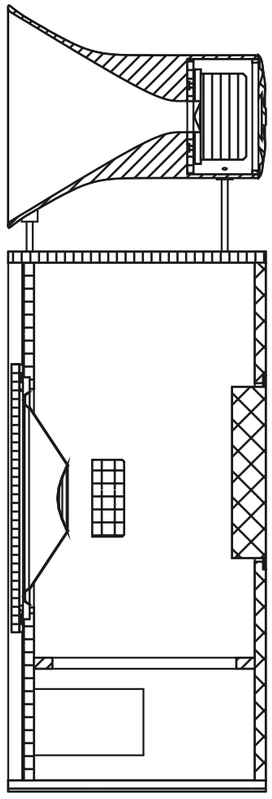
# General arrangement

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10. Section A-A (sheet 1): the product cut at x 280.0 and seen from the right; cut faces hatched, a different angle for each part.



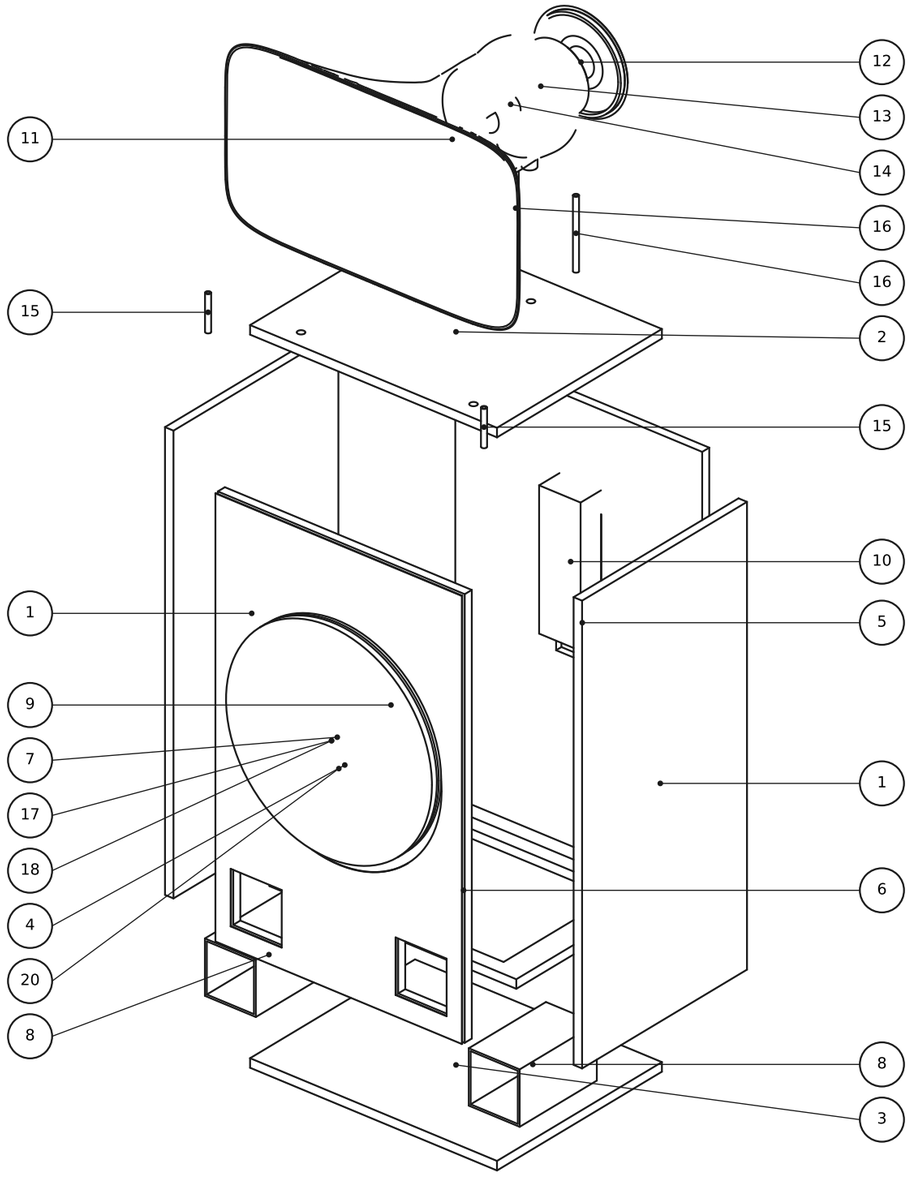
## Section A-A

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).




# Exploded view

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

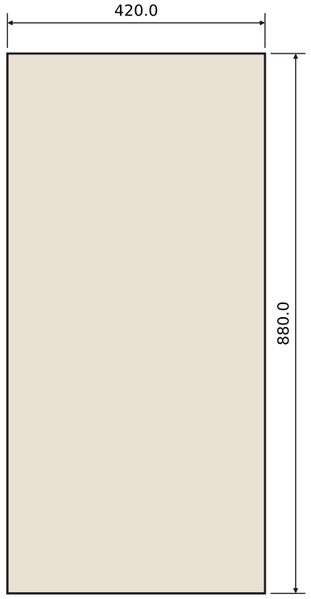
mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

**3 of 14**

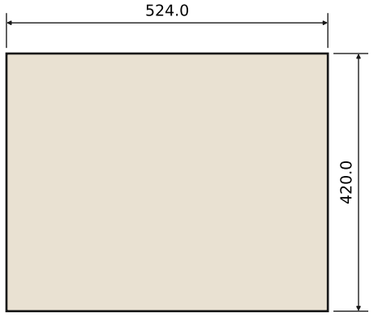
BAR-003 rev A

Scale 1:10. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.




**left, right**

18 mm baltic-birch, 4 for the set




**bottom**

18 mm baltic-birch, 2 for the set

# Sheet parts: plain panels

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

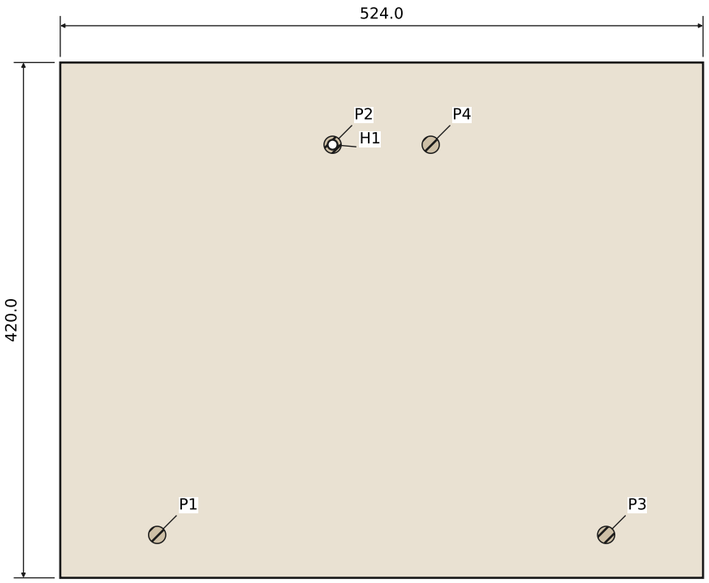
mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/top.dxf. Marks listed at the right.

| mark | size | x | y | depth |
|---|---|---|---|---|
| P1 | Ø14.0 | 79.0 | 35.0 | 15 from face |
| P2 | Ø14.0 | 222.0 | 353.0 | 15 from face |
| P3 | Ø14.0 | 445.0 | 35.0 | 15 from face |
| P4 | Ø14.0 | 302.0 | 353.0 | 15 from face |
| H1 | Ø8.0 | 222.0 | 353.0 | through |

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.




**top: 18 mm baltic-birch, 2 for the set, seen from its outer face**

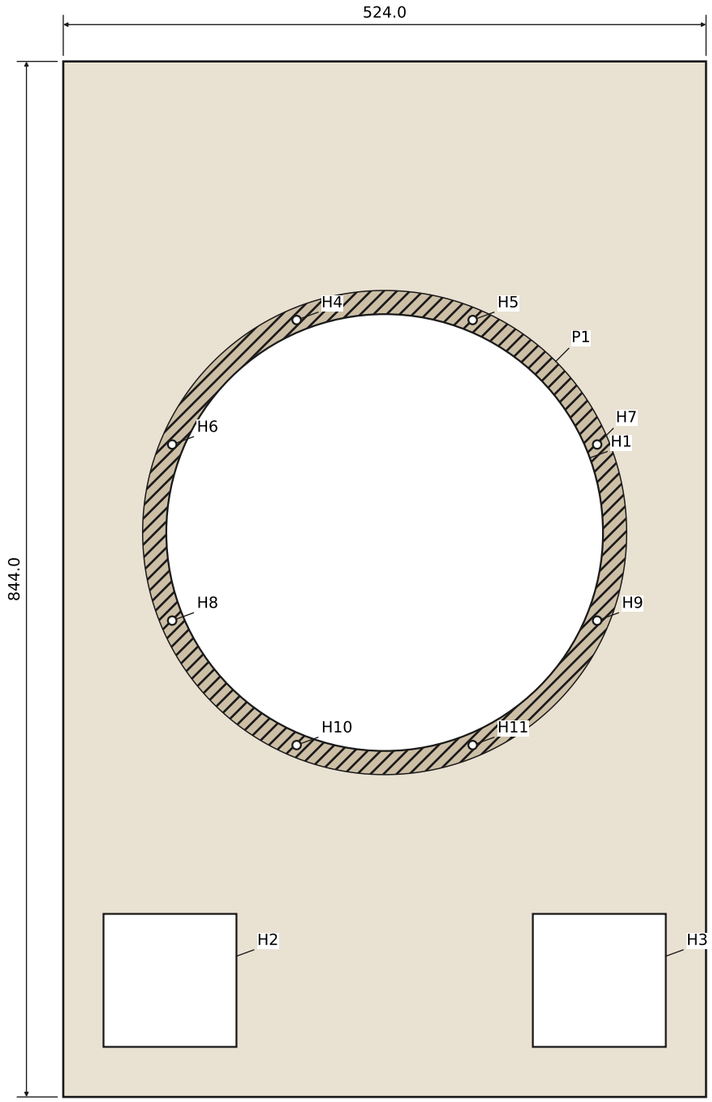
## Sheet part: top

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/front-panel.dxf. Marks listed at the right.




**front-panel: 18 mm baltic-birch, 2 for the set, seen from its outer face**

| mark | size | x | y | depth |
|---|---|---|---|---|
| P1 | Ø394.6 | 262.0 | 460.0 | 11 from face |
| H1 | Ø356.0 | 262.0 | 460.0 | through |
| H2 | 108.4 x 108.4 | 87.0 | 95.0 | through |
| H3 | 108.4 x 108.4 | 437.0 | 95.0 | through |
| H4 | Ø7.0 | 190.2 | 633.2 | through |
| H5 | Ø7.0 | 333.8 | 633.2 | through |
| H6 | Ø7.0 | 88.8 | 531.8 | through |
| H7 | Ø7.0 | 435.2 | 531.8 | through |
| H8 | Ø7.0 | 88.8 | 388.2 | through |
| H9 | Ø7.0 | 435.2 | 388.2 | through |
| H10 | Ø7.0 | 190.2 | 286.8 | through |
| H11 | Ø7.0 | 333.8 | 286.8 | through |

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

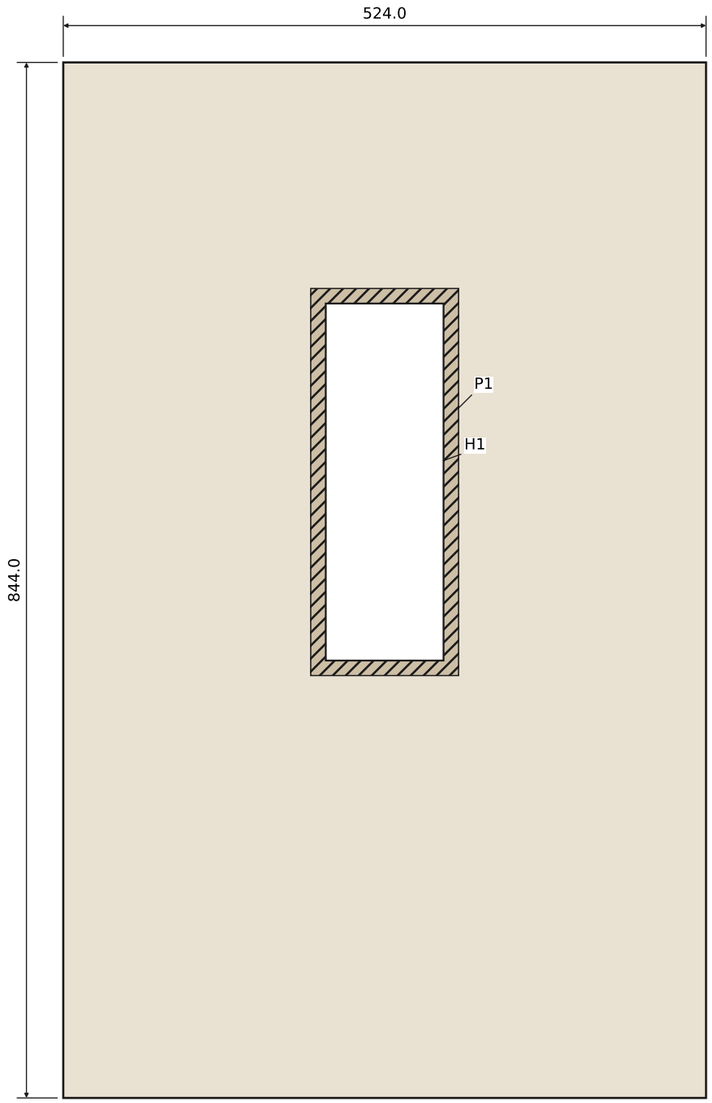
## Sheet part: front-panel

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/back.dxf. Marks listed at the right.




**back: 18 mm baltic-birch, 2 for the set, seen from its outer face**

| mark | size | x | y | depth |
|---|---|---|---|---|
| P1 | 120.5 x 315.5 | 262.0 | 502.0 | 4.5 from face |
| H1 | 96.0 x 291.0 | 262.0 | 502.0 | through |

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

## Sheet part: back

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

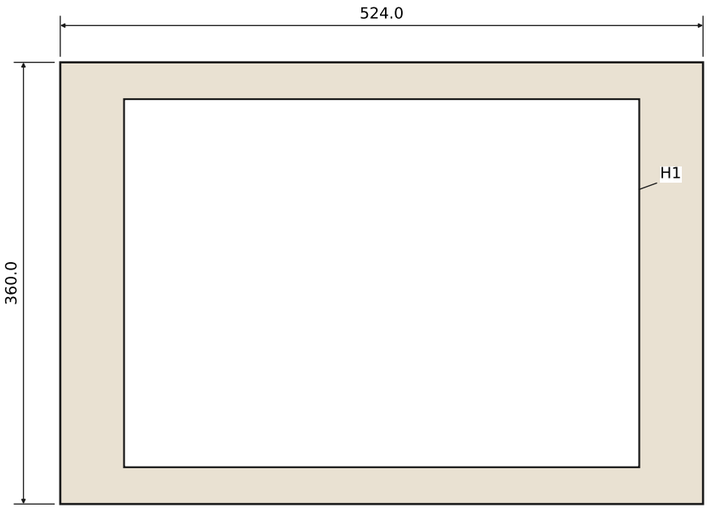
mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/brace.dxf. Marks listed at the right.

| mark | size | x | y | depth |
|---|---|---|---|---|
| H1 | 420.0 x 300.0 | 262.0 | 180.0 | through |

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.




**brace: 18 mm baltic-birch, 2 for the set, seen from its outer face**

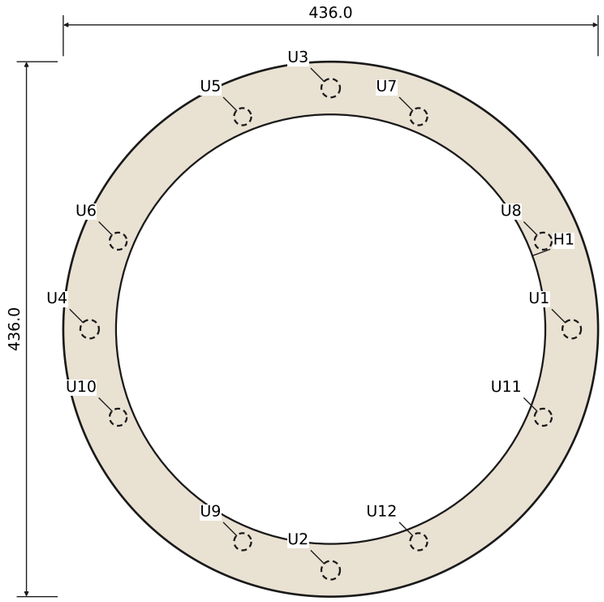
## Sheet part: brace

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/woofer-ring.dxf. Marks listed at the right.




**woofer-ring: 6 mm mdf, 2 for the set, seen from its outer face**

inner-face pockets: dxf/woofer-ring-underside.dxf (seen from that face)

| mark | size | x | y | depth |
|---|---|---|---|---|
| H1 | Ø350.0 | 218.0 | 218.0 | through |
| U1 | Ø15.2 | 414.5 | 218.0 | 3.2 from inside |
| U2 | Ø15.2 | 218.0 | 21.5 | 3.2 from inside |
| U3 | Ø15.2 | 218.0 | 414.5 | 3.2 from inside |
| U4 | Ø15.2 | 21.5 | 218.0 | 3.2 from inside |
| U5 | Ø14.0 | 146.3 | 391.2 | 4 from inside |
| U6 | Ø14.0 | 44.8 | 289.8 | 4 from inside |
| U7 | Ø14.0 | 289.8 | 391.2 | 4 from inside |
| U8 | Ø14.0 | 391.2 | 289.8 | 4 from inside |
| U9 | Ø14.0 | 146.3 | 44.8 | 4 from inside |
| U10 | Ø14.0 | 44.8 | 146.2 | 4 from inside |
| U11 | Ø14.0 | 391.2 | 146.2 | 4 from inside |
| U12 | Ø14.0 | 289.8 | 44.8 | 4 from inside |

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

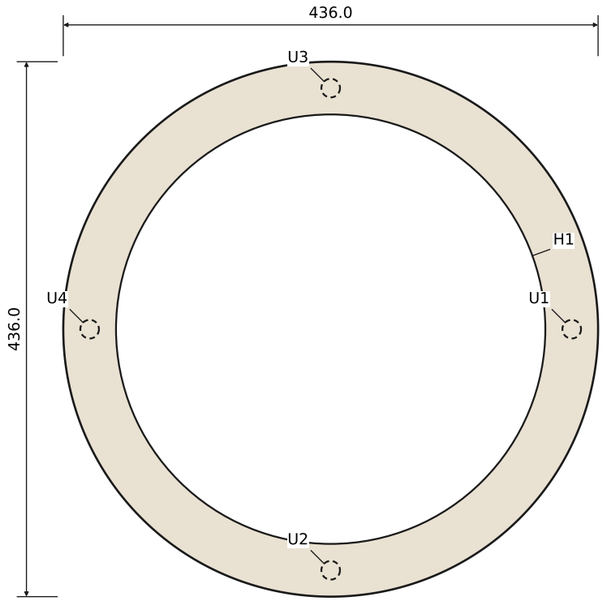
## Sheet part: woofer-ring

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/disc-frame.dxf. Marks listed at the right.




| mark | size | x | y | depth |
|---|---|---|---|---|
| H1 | Ø350.0 | 218.0 | 218.0 | through |
| U1 | Ø15.2 | 414.5 | 218.0 | 3.2 from inside |
| U2 | Ø15.2 | 218.0 | 21.5 | 3.2 from inside |
| U3 | Ø15.2 | 218.0 | 414.5 | 3.2 from inside |
| U4 | Ø15.2 | 21.5 | 218.0 | 3.2 from inside |

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

**disc-frame: 12 mm mdf, 2 for the set, seen from its outer face**

inner-face pockets: dxf/disc-frame-underside.dxf (seen from that face)

## Sheet part: disc-frame

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

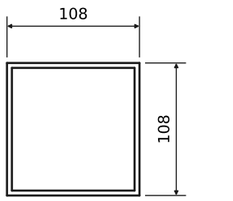
mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

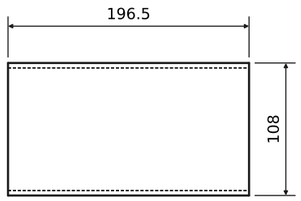
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

**8. port-left, port-right: asa print, on end; 327 cm3 each; 4 for the set; 1:5**

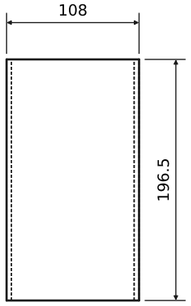
100 square, 196 long from the panel's face: the 2 together tune the box to 38 Hz; print 15 longer and trim to tune




**FRONT**



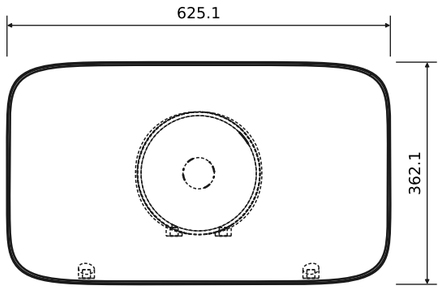

**FROM THE RIGHT**



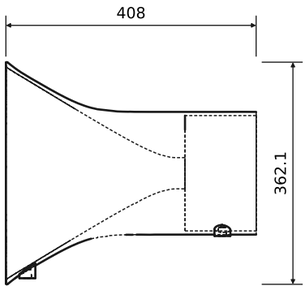

**FROM ABOVE**

**11. waveguide: cast, cast-resin; 5221 cm3 each; 2 for the set; 1:10**

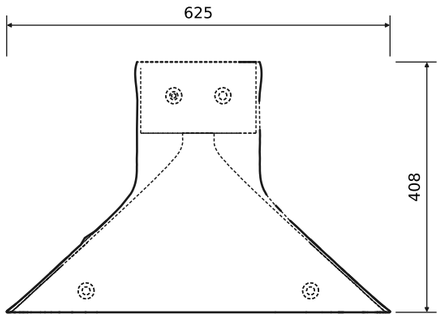
mouth 610 x 346, 278 deep, 2 in throat; a 10 rim rounded right round, 625 x 362 outside; a shell 6 thick following its flare; the driver in a pod 200 across behind, closed by a cap; four pads under it take the posts' brass inserts; cast in tinted clear resin in a two-part silicone mould from a CNC-cut plug, its faces polished satin (a slow deep-pour resin: the neck round the throat is solid, 5.2 L in all), or pigmented the same way for the opaque and marbled finishes




**FRONT**




**FROM THE RIGHT**




**FROM ABOVE**

## Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

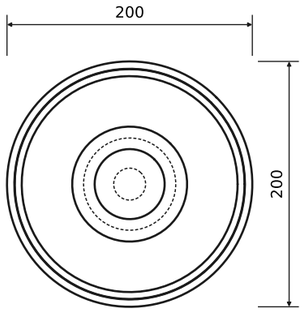
mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

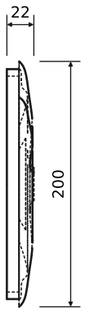
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

**12. waveguide-cap: cast, cast-resin; 187 cm3 each; 2 for the set; 1:5**

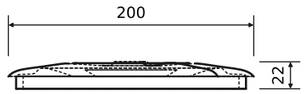
closes the pod behind the driver; a spigot locates it, three M4 screws hold it; the cable leaves through a post below




FRONT



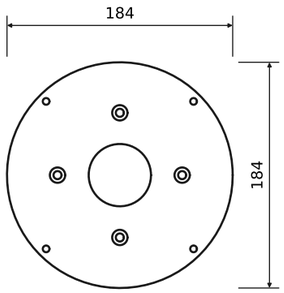

FROM THE RIGHT



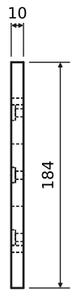

FROM ABOVE

**14. hf-adaptor: lathe, aluminium; 242 cm3 each; 2 for the set; 1:5**

184 across, 10 thick, bored 50.8 (the throat); 4 countersunk M6 on 101.6 take the driver from the front; 4 x 5.5 on 170 (at 45 degrees to them) take M5 socket caps into the brass inserts in the horn's boss; black anodised




FRONT




FROM THE RIGHT




FROM ABOVE

## Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

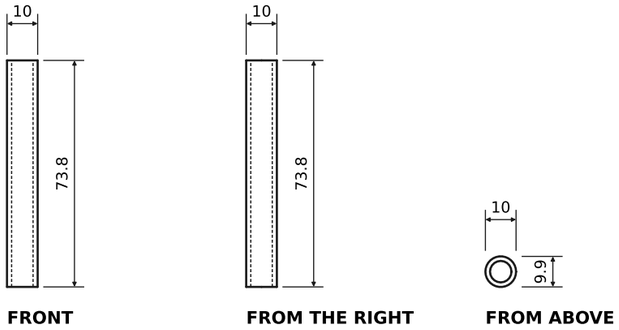
mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

**15. post-1, post-2: lathe, stainless; 3 cm3 each; 4 for the set; 1:2**

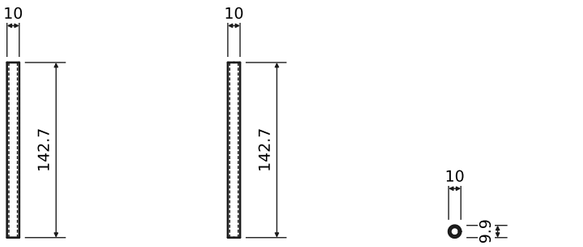
74 long (49 showing): stainless tube 10 x 7, threaded M10 x 1 12 at the top and 13 at the foot, each end into a brass insert




**FRONT** **FROM THE RIGHT** **FROM ABOVE**

**16. post-3, post-4: lathe, stainless; 6 cm3 each; 4 for the set; 1:5**

143 long (118 showing): stainless tube 10 x 7, threaded M10 x 1 12 at the top and 13 at the foot, each end into a brass insert; the HF cable runs down inside it




**FRONT** **FROM THE RIGHT** **FROM ABOVE**

## Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

| no. | item | make | qty | what | size | each USD |
|---|---|---|---|---|---|---|
| 1 | left, right | cut | 4 | 18 mm Baltic birch plywood, BB/BB | 420 x 880 | 26.80 |
| 2 | top | cut | 2 | 18 mm Baltic birch plywood, BB/BB | 524 x 420 | 15.96 |
| 3 | bottom | cut | 2 | 18 mm Baltic birch plywood, BB/BB | 524 x 420 | 15.96 |
| 4 | front-panel | cut | 2 | 18 mm Baltic birch plywood, BB/BB | 524 x 844 | 32.06 |
| 5 | back | cut | 2 | 18 mm Baltic birch plywood, BB/BB | 524 x 844 | 32.06 |
| 6 | brace | cut | 2 | 18 mm Baltic birch plywood, BB/BB | 524 x 360 | 13.68 |
| 7 | woofer-ring | cut | 2 | 6 mm MDF | 436 x 436 | 4.28 |
| 8 | port-left, port-right | print | 4 | ASA, FDM (outdoor, paintable) | 327 cm3 | 30.89 |
| 9 | woofer | buy | 2 | FaitalPRO 15PR400-8 15 in woofer (treated paper cone, neodymium, 8 ohm) | 393 frame, 356 cut-out, 165 deep | 194.00 |
| 10 | amp-plate | buy | 2 | Hypex FusionAmp FA122 DSP plate amplifier, 2 x 125 W into 4 ohm | 315 x 120 x 55 plate, 291 x 96 recess | 422.40 |
| 11 | waveguide | cast | 2 | Casting resin (polyurethane or epoxy), tinted clear, translucent or pigmented | 5221 cm3, 635 x 418 x 372 | 684.74 |
| 12 | waveguide-cap | cast | 2 | Casting resin (polyurethane or epoxy), tinted clear, translucent or pigmented | 187 cm3, 210 x 32 x 210 | 138.21 |
| 13 | hf | buy | 2 | B&C Speakers DE1085TN 2 in compression driver, titanium diaphragm, 100 mm voice coil, neodymium, 16 ohm (round 5: the owner's bigger throat) | 154 across, 86 deep, 4 x M6 on 102 | 512.00 |
| 14 | hf-adaptor | lathe | 2 | Aluminium 6061 | 194 x 20 x 194 | 43.82 |
| 15 | post-1, post-2 | lathe | 4 | Stainless steel 304 | 20 x 20 x 84 | 16.01 |
| 16 | post-3, post-4 | lathe | 4 | Stainless steel 304 | 20 x 20 x 153 | 16.83 |
| 17 | disc-frame | cut | 2 | 12 mm MDF | 436 x 436 | 4.28 |
| 18 | cloth-disc | buy | 2 | Acoustically transparent speaker cloth (a knit sold for grilles), about 0.6 x 0.95 m a grille, in the finish's colour | | [PRICE] |
| 19 | cloth-disc-magnets | buy | 2 | Grille magnets: eight 15 x 3 neodymium discs (N42), four attracting pairs, a set | | [PRICE] |
| 20 | bark-face, bark | buy | 6 | Bark for laminating, a square metre: paper birch (Betula papyrifera) sheet from fallen or felled trees, or a smooth-barked gum's shed bark pressed flat (or a UV-printed veneer of it); approve a sample board first | | [PRICE] |
| 21 | woofer-fixings | buy | 2 | M6 x 25 button heads (A2) with M6 T-nuts, a set of 8 (the 15PR400's eight holes on 375) | | [PRICE] |
| 22 | hf-fixings | buy | 2 | The 2 in driver's fixings: four M6 x 16 countersunk screws (A2), the driver to its adaptor ring, and four M5 x 16 socket caps with washers, the ring to the brass inserts in the horn's boss, a set | | [PRICE] |

8 more lines: bom.md

## Notes

1. Bark on the shell (sides, top, bottom and back): real birch bark sheet (Betula papyrifera, from fallen or felled trees) laminated to the plywood with PVA in a vacuum bag and sealed with a matt waterborne lacquer, wrapped round the shell's front edges; for eucalyptus, the shed bark of a smooth-barked gum (spotted or snow gum) pressed flat and laminated the same way, or a UV-printed veneer of it. Approve a sample board first.
2. Glue-up: sides to the bottom, the brace, then the front panel (set back 24 mm), the back, then the top.
3. The woofer flush in its rebate on a gasket, eight M6 T-nuts from inside, the ring over its frame; the port pushed in dry, trimmed to tune (impedance minimum at fb), then bonded.
4. The waveguide stands on four posts, stainless lamp tube 10 x 7 threaded M10 x 1 at both ends, screwed into brass inserts bonded in pads cast under it and into inserts in the top; the HF cable runs down inside a rear post.
5. The front: the panel faced in bark (1 mm) round the ring and the port, and a cloth disc over the woofer, its MDF frame standing on the ring on four magnet pairs, its face 18.5 clear of the cone at rest.

# Parts list and notes

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15